SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	**Date:**	27-08-26

Code of Conduct

I **_M Poojitha Reddy - 192472352________** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ___**M Poojitha Reddy______________________**

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4

Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as "best hotels near…" and "book a flight from Chennai to…". The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as "give the book to John," "sleep," and "put the file on the desk." The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1	4	4	4	3	4	19	95
2	4	4	4	3	4	19	
3	4	4	4	3	4	19	
4	4	4	4	3	4	19	
5	4	4	4	3	4	19	

Faculty In-charge	Course Coordinator	Program Director
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Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

1. Constituency Parsing vs. Dependency Parsing

Constituency parsing represents a sentence as a hierarchical tree of phrases such as noun phrases (NP), verb phrases (VP), and prepositional phrases (PP). It focuses on how words combine to form larger grammatical units. This representation is useful for understanding the overall syntactic structure and nested relationships in a sentence.

Dependency parsing, on the other hand, represents relationships directly between individual words. Each word depends on another word, usually with the main verb acting as the root. For example, in "The student solved the problem," dependency parsing directly identifies "student" as the subject of "solved" and "problem" as its object.

For conversational text, **dependency parsing is generally more suitable** for extracting meaningful relationships because it directly captures word-to-word relations such as subject, object, modifier, and location. This is especially useful for information extraction, question answering, and conversational AI, where identifying who did what to whom is important. Constituency parsing provides richer hierarchical information, but dependency parsing is often more compact and practical for relationship extraction.

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2. Top-Down Parsing vs. Earley Parsing

Conventional top-down parsing begins with the start symbol of the grammar and attempts to generate the input sentence by expanding grammar rules. It works well when the input is complete and the grammar is relatively simple. However, incomplete inputs such as "best hotels near…" can cause difficulties because the parser may expect the remaining words before determining whether the partial structure is valid.

Earley parsing is a dynamic-programming parsing algorithm that can handle context-free grammars and is particularly effective for incomplete and ambiguous input. It maintains possible parsing states as the input is processed, allowing it to represent multiple interpretations without immediately committing to one.

For interactive NLP applications, **Earley parsing is more appropriate**. Search queries and voice-assistant commands are often incomplete, continuously updated, or ambiguous. Earley parsing can maintain partial interpretations and efficiently predict what structures or words may follow. Therefore, it provides greater flexibility for real-time systems such as autocomplete, conversational search, and interactive assistants.

3. CFG vs. PCFG vs. Neural Constituency Parsers

A **Context-Free Grammar (CFG)** represents syntactic structure using explicit production rules such as S → NP VP. If a sentence has multiple valid parse trees, a CFG can represent all of them, but it does not inherently determine which interpretation is most likely. Thus, ambiguity is represented but not effectively resolved.

A **Probabilistic Context-Free Grammar (PCFG)** extends CFG by assigning probabilities to grammar rules. When multiple parse trees are possible, the parser can select the tree with the highest probability. This allows common grammatical structures to be preferred over less likely ones while retaining interpretable grammar rules.

Neural constituency parsers use machine-learning models, often based on transformer architectures, to learn syntactic patterns from large datasets. They can handle complex and ambiguous sentences more accurately because they learn contextual information and linguistic patterns from data. However, their decisions are less interpretable than explicit CFG or PCFG rules.

For the best balance between **interpretability and practical accuracy, PCFG is a strong choice**. It preserves explicit grammatical rules while using probabilities to resolve ambiguity. Neural parsers may achieve higher accuracy in many real-world settings, but they provide less transparent explanations. Therefore, PCFG is preferable when both understandable parsing decisions and effective ambiguity resolution are important.

4. Feature Structures vs. Traditional CFG

A traditional **Context-Free Grammar** mainly describes how grammatical categories can be combined. For example, rules may specify that a noun phrase can occur with a verb phrase. However, basic CFG rules do not naturally enforce detailed grammatical properties such as number, gender, person, or tense.

Feature structures extend grammar representations by attaching attributes and values to linguistic units.

For example, a noun phrase may contain features such as:
[number: plural, gender: feminine, person: third]

A verb can also contain corresponding features. During parsing, **feature unification** checks whether the features of related words are compatible. For example, a singular subject should normally combine with a singular verb, while grammatical gender agreement can be enforced in languages where gender affects agreement.

Feature structures are particularly useful in **multilingual NLP** because different languages have different agreement and grammatical constraints. They allow a grammar to represent these properties systematically rather than creating a large number of separate CFG rules. Therefore, feature structures improve grammatical accuracy, reduce rule complexity, and make NLP systems more flexible across languages.

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5. Subcategorization Frames

Subcategorization frames describe the types and number of arguments that a verb requires or permits. Different verbs have different argument requirements.

For example:

- **"give"** generally requires a giver, an object, and a recipient:

  give + NP + NP/PP
- **"sleep"** is typically intransitive and does not require a direct object:

  sleep
- **"put"** requires an object and a location:

  put + NP + PP

This information helps a voice assistant determine whether a sentence has a valid syntactic structure. For example, "She put the file" is incomplete because the verb *put* normally requires a location, while "She slept the book" is structurally inappropriate because *sleep* does not normally take a direct object.

Subcategorization is important for both **syntactic and semantic analysis**. Syntactically, it helps identify the grammatical roles of arguments. Semantically, it helps determine relationships such as agent, object, recipient, and location. Consequently, subcategorization frames allow voice assistants to interpret commands more accurately and distinguish between different meanings and argument structures.